PROBLEM SET 9 DUE 5.31

1. Prove the following statements:

(1) If A is a PID and $S \subseteq A\backslash\{0\}$ is a multiplicative subset, then $S^{-1}A$ is a PID and the prime elements of $S^{-1}A$ are the prime elements of A not dividing any element of S.

(2) If A is a UFD and $S \subseteq A\backslash\{0\}$ is a multiplicative subset, then $S^{-1}A$ is a UFD and the prime elements of $S^{-1}A$ are the prime elements of A not dividing any element of S.

2. Let A be a ring, $S \subseteq A$ a multiplicative subset, and let Σ be the set of ideals of A which do not meet S. Show that the maximal elements of Σ are in one-to-one correspondence with the maximal ideals of $S^{-1}A$. More specifically, the set of maximal elements in Σ is given by

$$\left\{\mathfrak{n}^c : \mathfrak{n} \in \operatorname{Specm} S^{-1}A\right\}.$$

3. Let A be a ring. A multiplicative subset S of A is said to be **saturated** if

$$xy \in S \quad \Longrightarrow \quad x \in S \text{ and } y \in S.$$

Show that the following are equivalent:

(i) S is saturated.
(ii) $A - S$ is a union of ideals.
(iii) $A - S$ is a union of prime ideals.

4. Let A be a ring and S be a multiplicative subset of A. Show that there is a smallest saturated multiplicative subset $\bar{S}$ of A containing S. We call $\bar{S}$ the **saturation** of S. Show that $\bar{S}$ is equal to any of the following

(i) $\left\{a \in A \,\middle|\, \exists a' \in A \text{ s.t. } aa' \in S\right\}$.

(ii) $\left\{a \in A \,\middle|\, \frac{a}{1} \text{ is a unit in } S^{-1}A\right\}$.

(iii) the complement of the union of ideals of A which do not meet S.

(iv) the complement of the union of prime ideals of A which do not meet S.

5. Let A be a ring and let S, T be two multiplicative subsets of A. Show that $S^{-1}A$ and $T^{-1}A$ are isomorphic as A-algebras if and only if $\bar{S} = \bar{T}$.

6. Let A be a ring. Show that every minimal prime ideal $\mathfrak{p}$ of A consists of zero-divisors.

Hint: $\operatorname{nil}(A_{\mathfrak{p}}) = \mathfrak{p}A_{\mathfrak{p}}$.

7. Let A be a ring and $S \subseteq A$ be a multiplicative subset.

(1) Suppose that M is an A-module such that for any $s \in S$, the homothety

$$h_s : M \to M; \quad x \mapsto sx$$

is bijective. Show that M can be made into an $S^{-1}A$-module by setting

$$\frac{a}{s}x \coloneqq h_s^{-1}(ax).$$

Show that this is the unique $S^{-1}A$-module structure on M such that the A-module structure derived from it by restriction of scalars coincides with the one given initially.

(2) Let N, N' be two $S^{-1}A$-modules and let $f : N \to N'$ be a map. Show that f is an $S^{-1}A$-module homomorphism if and only if f is an A-module homomorphism.

8. Let A be a ring, $S \subseteq A$ a multiplicative subset, and N an $S^{-1}A$-module. Show that N is flat as an $S^{-1}A$-module if and only if N is flat as an A-module.

9. Let A be an integral domain, K be its field of fractions, and M be an A-module.
Prove the following statements:

(1) M is torsion-free if and only if M can be embedded into a K-vector space.
(2) If M is torsion-free and finitely generated, then M can be embedded into a finitely generated free A-module.

10. Let $f : A \to B$ be a ring map, $S \subseteq A$ be a multiplicative subset, and N be a B-module. Put

$$T \coloneqq f(S)$$

Obviously, T is a multiplicative subset of B.

Show that there exists an isomorphism of $S^{-1}A$-modules

$$\begin{aligned} S^{-1}N &\cong T^{-1}N; \\ \frac{y}{s} &\mapsto \frac{y}{f(s)}. \end{aligned}$$

In particular, we have

$$S^{-1}B \cong T^{-1}B.$$

11. Simplify

$$\big(k[x,y]/(xy)\big)_{\bar{x}}$$

where k is a field and $\big(k[x,y]/(xy)\big)_{\bar{x}}$ is the localization of $k[x,y]/(xy)$ at the multiplicative subset generated by $\bar{x}$.